

EX.NO:	Creates Alarm Clock
DATE: / /2025	

AIM:

To develop a mobile application that allows the user to set an alarm using a TimePicker and triggers a notification at the selected time using AlarmManager.

ALGORITHM:

- Step-1: Start the Android Studio project and create a new mobile application.
- Step-2: Design the user interface by adding a TimePicker and a Set Alarm button.
- Step-3: Create a BroadcastReceiver class to handle the alarm event when the time is reached.
- Step-4: Get the selected hour and minute from the TimePicker when the user taps the Set Alarm button.
- Step-5: Create a Calendar object and set the selected hour and minute in it.
- Step-6: Create an Intent to call the BroadcastReceiver when the alarm triggers.
- Step-7: Create a PendingIntent using the Intent to schedule the alarm.
- Step-8: Use AlarmManager to set the alarm for the selected time.
- Step-9: When the alarm time is reached, the BroadcastReceiver displays a message indicating the alarm.
- Step-10: Stop the application.

SOURCE CODE:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    android:orientation="vertical">

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner" />

    <Button
        android:id="@+id/btnSetAlarm"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Set Alarm"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

MainActivity.Java:

```
package com.example.alarmclock;

import androidx.appcompat.app.AppCompatActivity;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TimePicker;
```

```
import android.widget.Toast;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    TimePicker timePicker;
    Button btnSetAlarm;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        timePicker = findViewById(R.id.timePicker);
        btnSetAlarm = findViewById(R.id.btnSetAlarm);

        btnSetAlarm.setOnClickListener(v -> setAlarm());
    }

    private void setAlarm() {
        int hour = timePicker.getHour();
        int minute = timePicker.getMinute();

        Calendar calendar = Calendar.getInstance();
        calendar.set(Calendar.HOUR_OF_DAY, hour);
        calendar.set(Calendar.MINUTE, minute);
        calendar.set(Calendar.SECOND, 0);

        Intent intent = new Intent(MainActivity.this, AlarmReceiver.class);
        PendingIntent pendingIntent = PendingIntent.getBroadcast(
            this, 0, intent, PendingIntent.FLAG_IMMUTABLE);

        AlarmManager alarmManager = (AlarmManager) getSystemService(ALARM_SERVICE);

        alarmManager.setExact(
            AlarmManager.RTC_WAKEUP,
```

```

        calendar.getTimeInMillis(),
        pendingIntent
    );

    Toast.makeText(this, "Alarm Set!", Toast.LENGTH_SHORT).show();
}
}

```

AndroidManifest.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Alarmclock">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:theme="@style/Theme.Alarmclock">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <receiver android:name=".AlarmReceiver" />
    </application>

```

</application>

</manifest>

AlarmReceiver.Java:

```
package com.example.alarmclock;
```

```
import android.content.BroadcastReceiver;
```

```
import android.content.Context;
```

```
import android.content.Intent;
```

```
import android.widget.Toast;
```

```
public class AlarmReceiver extends BroadcastReceiver {
```

```
    @Override
```

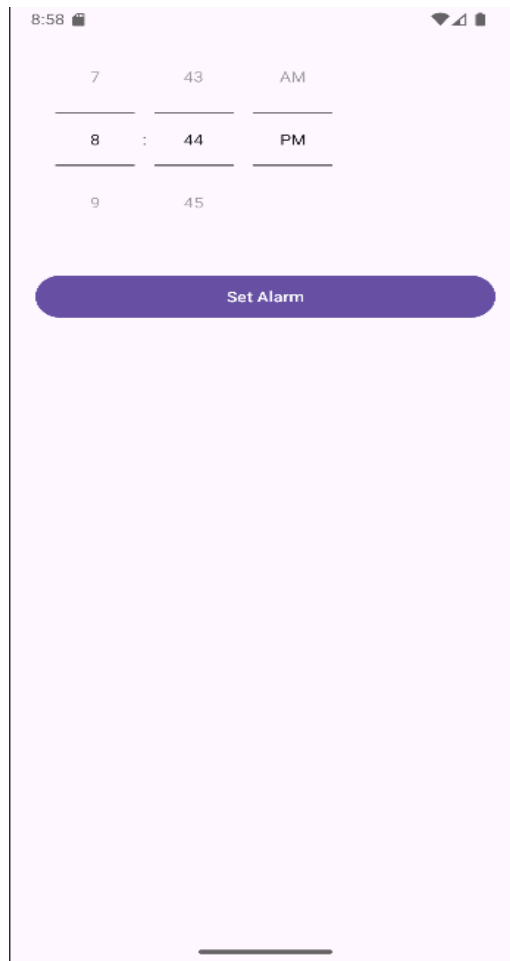
```
    public void onReceive(Context context, Intent intent) {
```

```
        Toast.makeText(context, "Alarm!", Toast.LENGTH_SHORT).show();
```

```
    }
```

```
}
```

OUTPUT:



RESULT:

The mobile application for creating an alarm clock was successfully developed.